

KTrade v10.3

Run Guide, File Explanation, and Finnhub Update

Project folder: C:\trading-agent\Ktrade_v10_3

This PDF now includes the latest Finnhub integration changes, daily run commands, file usage, and before/after explanation.

Area	What This PDF Covers
Daily Run	Backend, dashboard, scanner, paper trading flow
Backtesting	Daily VectorBT and intraday VectorBT commands
v10.3 Files	Backend, frontend, scanner, data, risk, tests, docs
Finnhub Update	New quote/news/candle fallback support and commands

Part 1 - Daily Run Guide

?KTrade v10.3 - Daily Run Guide

Project Folder

C:\trading-agent\Ktrade_v10_3

Main Dashboard URL

```
http://localhost:5001
```

Important Note

For normal daily use, you only need:

1. Backend / Dashboard server
2. Scanner

VectorBT is NOT required every day. Use VectorBT only when you want to refresh or retest strategy/backtest parameters.

1) Start Backend / Dashboard Server

Open CMD window 1 and run:

```
cd /d C:\trading-agent\Ktrade_v10_3
.venv\Scripts\python.exe backend\ktrade_alpaca.py
```

Use:

- Starts the local dashboard server.
- Connects to Alpaca paper account.
- Handles positions, account, orders, buy/sell paper orders.
- Handles broker reconciliation.
- Keeps the web app running at http://localhost:5001

Keep this CMD window open while using the dashboard.

2) Open Dashboard

Open browser and go to:

```
http://localhost:5001
```

Use:

- Shows signals, positions, trades, alerts, performance, AI advisor.
- Shows Alpaca paper account information.
- Shows Broker OK / REVIEW reconciliation status.

3) Run Daily / Swing Scanner

Open CMD window 2 and run:

```
cd /d C:\trading-agent\Ktrade_v10_3
set KTRADE_DATA_PROVIDER=yfinance
set KTRADE_SCAN_SYMBOLS=
set KTRADE_SCAN_UNIVERSE=extended
set KTRADE_SCAN_INTERVAL=1d
.venv\Scripts\python.exe agent\ktrade_agent_v9.py --score-only
```

Use:

- Scans stocks using daily candles.
- Best for daily / swing / momentum signals.
- Updates dashboard signals.
- Does not place orders automatically.
- Saves result here:

C:\trading-agent\Ktrade_v10_3\data\ktrade_scan_latest.json

Meaning of settings:

- KTRADE_DATA_PROVIDER=yfinance means use yfinance as data source.
- KTRADE_SCAN_SYMBOLS= blank means use default/extended universe.
- KTRADE_SCAN_UNIVERSE=extended means scan your full extended stock list.
- KTRADE_SCAN_INTERVAL=1d means daily candles.
- --score-only means scanner only creates signals and does not trade.

4) Run Intraday Scanner

Use this only when you want intraday signals.

```
cd /d C:\trading-agent\Ktrade_v10_3
set KTRADE_DATA_PROVIDER=yfinance
set KTRADE_SCAN_SYMBOLS=
set KTRADE_SCAN_UNIVERSE=extended
set KTRADE_SCAN_INTERVAL=5m
.venv\Scripts\python.exe agent\ktrade_agent_v9.py --score-only
```

Use:

- Scans stocks using 5-minute candles.
- Best for intraday trade checking.
- Updates dashboard with intraday-style signals.
- Does not place orders automatically.

Simple rule:

- 1d = daily / swing / momentum style
- 5m = intraday style

5) Run Tests

Optional. Run when you want to confirm project health.

```
cd /d C:\trading-agent\Ktrade_v10_3
.venv\Scripts\python.exe test_v103_fixes.py
.venv\Scripts\python.exe data\test_price_sanity.py
```

Use:

- Checks v10.3 safety fixes.
- Checks bad-price / decimal-shift protection.
- Checks risk, strategy, execution behavior.

Expected result:

- v10.3 tests should show passed.
- Price sanity tests should show ALL PRICE-SANITY TESTS PASSED.

6) Run VectorBT Backtest

Not needed every day.

Run only when you want to refresh backtest/approved parameters.

```
cd /d C:\trading-agent\Ktrade_v10_3  
.venv\Scripts\python.exe ktrade_vectorbt.py
```

Use:

- Runs historical backtest.
- Updates approved strategy parameters.
- Helps check whether strategy worked historically.

7) Run Intraday VectorBT Backtest

Optional. Use when you want intraday strategy backtest.

```
cd /d C:\trading-agent\Ktrade_v10_3  
.venv\Scripts\python.exe ktrade_intraday_vectorbt.py --fast --universe extended
```

Use:

- Runs intraday VectorBT backtest.
- Creates intraday approved parameters/report.

Daily Minimum Commands

Cmd 1:

```
cd /d C:\trading-agent\Ktrade_v10_3  
.venv\Scripts\python.exe backend\ktrade_alpaca.py
```

Cmd 2:

```
cd /d C:\trading-agent\Ktrade_v10_3  
set KTRADE_DATA_PROVIDER=yfinance  
set KTRADE_SCAN_SYMBOLS=  
set KTRADE_SCAN_UNIVERSE=extended  
set KTRADE_SCAN_INTERVAL=1d  
.venv\Scripts\python.exe agent\ktrade_agent_v9.py --score-only
```

Then open:

```
http://localhost:5001
```

Easy Double-Click Files

These files are also available in this folder:

```
START_BACKEND.cmd  
RUN_SCANNER_DAILY.cmd  
RUN_SCANNER_INTRADAY.cmd  
RUN_VECTORBT.cmd  
RUN_INTRADAY_VECTORBT.cmd  
RUN_TESTS.cmd
```

You can double-click them instead of typing commands.

Which Files Are Actually Run-

Main runtime files:

- backend\ktrade_alpaca.py
- agent\ktrade_agent_v9.py
- frontend\KTrade_preview.html

Support files used automatically:

- backend\broker_reconciler.py
- agent\broker_adapter.py
- data\ktrade_data.py
- data\price_sanity.py
- risk\ktrade_risk.py

Backtest files:

- ktrade_vectorbt.py
- ktrade_intraday_vectorbt.py

Test files:

- test_v103_fixes.py
- data\test_price_sanity.py

Safety Notes

- --score-only does not place orders.
- Paper orders are controlled by KTRADE_PAPER_ORDER_SUBMISSION in .env.
- This project is for Alpaca PAPER trading/testing, not live money trading.
- Broker reconciliation checks dashboard state against Alpaca broker truth.
- Bad price guard blocks decimal-shift/bad tick problems before trade approval.

Part 2 - v10.3 File Explanation and Changes

?KTrade v10.3 Changes and File Explanation

Project Folder

C:\trading-agent\Ktrade_v10_3

Purpose Of This Document

This document explains:

- What changed in v10.3
- Which files are new
- Which files were updated
- What each important file is used for
- What existed in the older version before v10.3

High-Level Summary

Before v10.3, the project already had:

- Dashboard UI
- Alpaca paper backend
- Scanner / signal engine
- Daily and intraday scanner support
- VectorBT backtesting
- Basic broker reconciliation
- Paper trade placement

v10.3 adds stronger safety and better trading correctness:

- Broker-truth reconciliation from real Alpaca fills
- Verified closed trades from Alpaca fill activity
- Bad-price / decimal-shift protection
- Safer execution: record fills only after broker confirms fill
- Broker adapter for paper bracket orders
- Better RiskEngine tracking of real positions/equity
- Improved scanner/strategy confirmation logic
- VectorBT look-ahead prevention
- Market-hours and holiday awareness

Main v10.3 Improvements

1. Broker Truth Reconciliation

Use:

- Makes Alpaca the source of truth.
- Compares dashboard/agent positions with real broker positions.
- Helps detect if the dashboard thinks a trade is open but Alpaca is flat.

Added/updated files:

- backend\broker_reconciler.py
- backend\ktrade_alpaca.py

Before v10.3:

- Project had simpler reconciliation.

- It could show positions/orders, but closed-trade P&L and broker fill truth were not as strong.

After v10.3:

- Real Alpaca fill history is used to reconstruct closed trades.
- Backend exposes broker truth endpoints.

Important endpoints:

- /reconciliation
- /reconcile_truth
- /closed_trades

2. Verified Closed Trades

Use:

- Builds closed trades from real Alpaca fill activities.
- Calculates P&L only from real broker fills.
- Helps avoid fake or estimated trade results.

Added/updated files:

- backend\broker_reconciler.py
- backend\ktrade_alpaca.py

Before v10.3:

- Dashboard could show positions and orders.
- Closed trade reporting was weaker or based more on dashboard/order state.

After v10.3:

- Closed trades are reconstructed from Alpaca fill activity.
- Win rate, profit factor, average win/loss can be based on actual fills.

3. Bad Price / Decimal Shift Guard

Use:

- Protects the strategy from wrong market prices.
- Example: if real price is around 236 but bad data says 2300, this guard blocks it.
- Prevents false momentum signals caused by bad ticks.

New files:

- data\price_sanity.py
- data\test_price_sanity.py

Updated files:

- data\ktrade_data.py
- agent\ktrade_agent_v9.py

Before v10.3:

- A bad market tick could enter the data pipeline.
- Strategy might treat a bad spike as strong momentum.

After v10.3:

- Bad ticks are rejected or carried forward using last-good price.
- RiskEngine also checks price again before approving a trade.

4. Safer Paper Execution

Use:

- Adds a broker adapter for Alpaca paper bracket orders.
- The agent records a position only after broker confirms a fill.
- Prevents phantom positions.

New file:

- agent\broker_adapter.py

Updated file:

- agent\ktrade_agent_v9.py

Before v10.3:

- Execution could simulate or assume fills more easily.
- There was more risk of dashboard/agent state drifting from broker truth.

After v10.3:

- Broker fill confirmation is required before recording a real trade fill.
- If an order is not filled, the agent does not record it as a real position.

5. Risk Engine Improvements

Use:

- Tracks real open positions.
- Syncs positions from broker truth.
- Updates equity from broker account.
- Enforces max positions and exposure more correctly.

Updated file:

- agent\ktrade_agent_v9.py

Before v10.3:

- Some risk checks could depend on internal state.
- If internal state was stale, risk checks could be less reliable.

After v10.3:

- RiskEngine can sync from broker positions.
- Daily loss guard can use broker equity.
- Position limits are more reliable.

6. Strategy Confirmation Fix

Use:

- Score alone is not enough.
- The strategy that produced the score must also confirm a signal.

Updated file:

- agent\ktrade_agent_v9.py

Before v10.3:

- A ticker could score well but the executable strategy signal might not fully confirm.

After v10.3:

- If score says MOMENTUM, momentum signal must confirm.
- If score says MACD_EMA, MACD/EMA signal must confirm.

- ORB is used only for intraday data.

7. ORB Intraday Fix

Use:

- ORB means Opening Range Breakout.
- It only makes sense on intraday candles, not daily candles.

Updated file:

- agent\ktrade_agent_v9.py

Before v10.3:

- ORB could influence score where it should not.

After v10.3:

- ORB only contributes on intraday data.
- Daily scoring weights are adjusted without ORB.

8. VectorBT Look-Ahead Fix

Use:

- Prevents backtest from using same-bar future-like information.
- Signal created on bar T is acted on at bar T+1.

Updated file:

- ktrade_vectorbt.py

Before v10.3:

- Backtest could be slightly too optimistic if signal and trade happened on same bar.

After v10.3:

- Entries and exits are shifted by one bar.
- Backtest becomes more realistic.

9. Market Hours / Holiday Awareness

Use:

- Helps the agent understand market timing better.
- Includes weekend and 2026 holiday checks.

Updated file:

- agent\ktrade_agent_v9.py

Before v10.3:

- Market phase logic was simpler.

After v10.3:

- Uses America/New_York market timing more carefully.
- Avoids trading-day mistakes on weekends/holidays.

10. v10.3 Test Suite

Use:

- Confirms safety fixes are working.
- Checks bad-price guard, broker fill behavior, risk logic, and strategy logic.

New file:

- test_v103_fixes.py

Before v10.3:

- No dedicated v10.3 safety test suite.

After v10.3:

- Test suite proves important safety fixes.
- Expected result: 22 passed, 0 failed.

Important File List

backend\ktrade_alpaca.py

Use:

- Main backend server.
- Starts dashboard at http://localhost:5001
- Connects to Alpaca paper account.
- Provides account, positions, prices, orders, buy/sell APIs.
- Provides broker reconciliation endpoints.

Before version:

- Basic Alpaca backend and dashboard bridge.

v10.3 version:

- Adds broker truth endpoints.
- Adds verified closed trades endpoint.
- Adds client_order_id support for traceable orders.

Run command:

```
cd /d C:\trading-agent\Ktrade_v10_3  
.\venv\Scripts\python.exe backend\ktrade_alpaca.py
```

backend\broker_reconciler.py

Use:

- Reconstructs closed trades from real Alpaca fill activities.
- Detects broker/agent position desync.
- Calculates verified P&L stats.

Before version:

- This file did not exist.

v10.3 version:

- New broker-truth reconciliation engine.

Do you run it directly-

- No. Backend imports it automatically.

agent\ktrade_agent_v9.py

Use:

- Main signal scanner and agent logic.
- Scores stocks.
- Applies strategy rules.
- Applies risk checks.

- Produces BUY/WATCH signals.

Before version:

- Scanner and agent existed.
- Had scoring, strategy, and risk logic.

v10.3 version:

- Adds better strategy confirmation.
- Adds bad-price hard gate.
- Adds real broker position/equity sync support.
- Adds broker-confirmed fill logic.
- Adds market-hours and holiday logic.
- ORB is intraday-only.

Run command:

```
cd /d C:\trading-agent\Ktrade_v10_3
set KTRADE_DATA_PROVIDER=yfinance
set KTRADE_SCAN_SYMBOLS=
set KTRADE_SCAN_UNIVERSE=extended
set KTRADE_SCAN_INTERVAL=1d
.venv\Scripts\python.exe agent\ktrade_agent_v9.py --score-only
```

agent\broker_adapter.py

Use:

- Connects agent execution logic to Alpaca paper backend.
- Submits paper bracket orders.
- Waits for broker fill confirmation.

Before version:

- This file did not exist.

v10.3 version:

- New adapter for safer paper execution.

Do you run it directly-

- No. Agent imports/uses it when wired.

data\ktrade_data.py

Use:

- Market data feed layer.
- Gets prices/history from Polygon, Alpaca, or yfinance depending on settings.
- Feeds scanner and strategy logic.

Before version:

- Data source/fallback logic existed.

v10.3 version:

- Adds price sanity guard into snapshot flow.
- Bad prices can be filtered before scanner sees them.

Do you run it directly-

- No. Agent imports it automatically.

data\price_sanity.py

Use:

- Detects bad prices, decimal-shift errors, non-positive prices, large suspicious jumps.
- Protects scanner and risk approval path.

Before version:

- This file did not exist.

v10.3 version:

- New bad-tick safety guard.

Do you run it directly-

- No. Used by data and agent files.

data\test_price_sanity.py

Use:

- Test file for price sanity guard.
- Confirms bad KLAC-style 10x price is blocked.

Before version:

- This file did not exist.

Run command:

```
cd /d C:\trading-agent\Ktrade_v10_3  
.\venv\Scripts\python.exe data\test_price_sanity.py
```

frontend\KTrade_preview.html

Use:

- Main dashboard UI.
- Shows signals, positions, trades, alerts, performance.
- Calls backend APIs.

Before version:

- Dashboard existed.

v10.3 version:

- Uses backend signals dynamically.
- Shows trade type/timeframe/hold/exit rule.
- Shows broker reconciliation status.

Do you run it directly-

- No. Backend serves it at <http://localhost:5001>

ktrade_vectorbt.py

Use:

- Daily/swing backtest engine.
- Tests strategy historically.
- Creates approved parameters.

Before version:

- VectorBT backtest existed.

v10.3 version:

- Adds one-bar signal shift to avoid look-ahead bias.

Run command:

```
cd /d C:\trading-agent\Ktrade_v10_3  
.venv\Scripts\python.exe ktrade_vectorbt.py
```

ktrade_intraday_vectorbt.py

Use:

- Intraday VectorBT backtest engine.
- Tests intraday strategies historically.

Before version:

- Already existed.

v10.3 version:

- No major change compared with your current copied version.

Run command:

```
cd /d C:\trading-agent\Ktrade_v10_3  
.venv\Scripts\python.exe ktrade_intraday_vectorbt.py --fast --universe extended
```

risk\ktrade_risk.py

Use:

- Risk rule definitions and risk guard helpers.

Before version:

- Already existed.

v10.3 version:

- Same or mostly unchanged compared with previous package.

Do you run it directly-

- No. Used by agent/risk logic.

test_v103_fixes.py

Use:

- v10.3 safety test suite.
- Confirms important fixes work.

Before version:

- This file did not exist.

Run command:

```
cd /d C:\trading-agent\Ktrade_v10_3  
.venv\Scripts\python.exe test_v103_fixes.py
```

Expected result:

- 22 passed, 0 failed.

Run Script Files

```
START_BACKEND.cmd
```

Use:

- Double-click shortcut to start backend.

Before version:

- Not in original zip. Added locally for your convenience.

```
RUN_SCANNER_DAILY.cmd
```

Use:

- Double-click shortcut to run daily scanner.

Before version:

- Added locally for your convenience.

```
RUN_SCANNER_INTRADAY.cmd
```

Use:

- Double-click shortcut to run intraday scanner.

Before version:

- Added locally for your convenience.

```
RUN_TESTS.cmd
```

Use:

- Double-click shortcut to run test_v103_fixes.py and data\test_price_sanity.py.

Before version:

- Added locally for your convenience.

```
RUN_COMMANDS_V10_3.txt
```

Use:

- Command reference document.
- Explains daily commands.

Before version:

- Added locally for your convenience.

KTrade_v10_3_Daily_Run_Guide.txt / .md

Use:

- Daily user guide with commands and explanations.

Before version:

- Added locally for your convenience.

What You Normally Run Daily

CMD 1: Backend

```
cd /d C:\trading-agent\Ktrade_v10_3
.venv\Scripts\python.exe backend\ktrade_alpaca.py
```

CMD 2: Daily scanner

```
cd /d C:\trading-agent\Ktrade_v10_3
set KTRADE_DATA_PROVIDER=yfinance
set KTRADE_SCAN_SYMBOLS=
set KTRADE_SCAN_UNIVERSE=extended
```

```
set KTRADE_SCAN_INTERVAL=1d
.venv\Scripts\python.exe agent\ktrade_agent_v9.py --score-only
```

Open dashboard:

```
http://localhost:5001
```

What You Run For Intraday

```
cd /d C:\trading-agent\Ktrade_v10_3
set KTRADE_DATA_PROVIDER=yfinance
set KTRADE_SCAN_SYMBOLS=
set KTRADE_SCAN_UNIVERSE=extended
set KTRADE_SCAN_INTERVAL=5m
.venv\Scripts\python.exe agent\ktrade_agent_v9.py --score-only
```

What You Do Not Need To Run Daily

Do not run daily unless needed:

- ktrade_vectorbt.py
- ktrade_intraday_vectorbt.py
- test_v103_fixes.py
- data\test_price_sanity.py

Use VectorBT only when:

- Strategy logic changed
- New stocks added
- You want fresh approved parameters
- You want fresh backtest report

Use tests only when:

- You updated files
- Something is not working
- Before Git push/deployment

Final Recommendation

Use v10.3 as your current testing version.

Keep the older project as backup.

Do not overwrite old project until v10.3 is tested for a few days.

Latest Update - Finnhub Integration

What was done now

Finnhub was added to the KTrade v10.3 project as an optional market-data, quote, and news source.

This does not replace Alpaca paper trading. Alpaca is still the broker truth for account, positions, orders, and paper trades.

Finnhub is used to improve quote/news support and provide another fallback when Polygon or Alpaca data has issues.

Files updated now

1. data\ktrade_data.py

Use: Main scanner/backtest market-data layer.

What changed: Added Finnhub API support for historical candles and quote snapshots.

Before: Data flow was mainly Polygon -> Alpaca -> yfinance.

Now: Data flow can be Polygon -> Alpaca -> Finnhub -> yfinance when provider is auto.

Direct mode: set KTRADE_DATA_PROVIDER=finnhub.

2. backend\ktrade_alpaca.py

Use: Backend server for dashboard, Alpaca paper trading, positions, orders, prices, and API routes.

What changed: Added Finnhub helper, quote fallback, and a news API endpoint.

New endpoint: <http://localhost:5001/news/AAPL>

Existing endpoint improved: <http://localhost:5001/quote/AAPL>

3. .env.template

Use: Example environment file.

What changed: Added FINNHUB_API_KEY example variable.

4. .env

Use: Your local secret settings.

What changed: Added FINNHUB_API_KEY placeholder if it was missing.

You must paste your real Finnhub key there.

5. RUN_SCANNER_FINNHUB_DAILY.cmd

Use: Runs the daily scanner using Finnhub as the data provider.

6. RUN_SCANNER_FINNHUB_INTRADAY.cmd

Use: Runs the intraday scanner using Finnhub as the data provider.

7. KTrade_v10_3_Finnhub_Integration_Guide.txt

Use: Separate short guide explaining Finnhub setup and commands.

How to configure Finnhub

Open this file:

C:\trading-agent\Ktrade_v10_3\.env

Add or update this line:

FINNHUB_API_KEY=your_real_finnhub_key_here

How to run with Finnhub daily scanner

Open CMD 1 for backend:

```
cd /d C:\trading-agent\Ktrade_v10_3
.venv\Scripts\python.exe backend\ktrade_alpaca.py
```

Open CMD 2 for Finnhub daily scanner:

```
cd /d C:\trading-agent\Ktrade_v10_3
RUN_SCANNER_FINNHUB_DAILY.cmd
```

Open dashboard:

<http://localhost:5001>

How to run with Finnhub intraday scanner

Open CMD 2 and run:

```
cd /d C:\trading-agent\Ktrade_v10_3
RUN_SCANNER_FINNHUB_INTRADAY.cmd
```

Important explanation

Alpaca = paper trading, broker account, positions, orders, source of truth.

Finnhub = quotes, news, optional candles, fallback market data.

Polygon = primary market data if working and plan allows it.

yfinance = final backup/fallback for testing and history.

So after this update, Finnhub helps the project get more data/news, but it does not place trades. Trades still go through Alpaca paper trading only.

Part 3 - Finnhub Integration Added Now

KTrade v10.3 - Finnhub Integration Guide

What was added

Finnhub is now integrated as an optional market-data and news source.

Updated files

1. data\ktrade_data.py

Use: scanner/backtest market data feed.

New: supports FINNHUB_API_KEY and KTRADE_DATA_PROVIDER=finnhub.

Flow: Polygon -> Alpaca -> Finnhub -> yfinance when KTRADE_DATA_PROVIDER=auto.

2. backend\ktrade_alpaca.py

Use: backend server and dashboard API.

New: /quote/<ticker> can use Finnhub fallback when Alpaca quote is missing.

New: /news/<ticker> returns Finnhub company news if FINNHUB_API_KEY is configured.

3. .env.template

Use: example environment settings.

New: FINNHUB_API_KEY placeholder and data-provider note.

4. .env

Use: your local secret settings.

New: FINNHUB_API_KEY= blank placeholder if it was missing. Put your key there.

```
5. RUN_SCANNER_FINNHUB_DAILY.cmd
```

Use: run daily scanner directly with Finnhub.

```
6. RUN_SCANNER_FINNHUB_INTRADAY.cmd
```

Use: run intraday scanner directly with Finnhub.

How to use

Step 1: Add your Finnhub key in:

C: rading-agent\Ktrade_v10_3\.env

Add/fill this line:

FINNHUB_API_KEY=your_real_key_here

Step 2: Start backend:

```
cd /d C: rading-agent\Ktrade_v10_3
```

```
.venv\Scripts\python.exe backend\ktrade_alpaca.py
```

Step 3: Run Finnhub daily scanner in another CMD:

```
cd /d C: rading-agent\Ktrade_v10_3
```

```
RUN_SCANNER_FINNHUB_DAILY.cmd
```

Step 4: Open dashboard:

```
http://localhost:5001
```

Important note

Finnhub is useful for quotes and news/sentiment. For broker truth and paper trades, Alpaca is still the source of truth.